

Pietro Girotto

MACHINE LEARNING ENGINEER

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Work Experience

Uber

Krakow, Poland

AUTOMATION SPECIALIST INTERN

Jun. 2024 - May 2024

- Deployed and maintained tools serving nations-wide EMEA markets.
- Executed end-to-end problem-solving projects mainly in **Python** and **SQL**, from requirements gathering to post-deployment support.
- Discovered and dealt with bugs in legacy code that were affecting as much as **40% of the user base**.
- Drove improvements for version control (**Git**) policy changes in my department.
- Achieved **exclusively maximum ratings** in stakeholders' satisfaction, delivery time and communication.

Education

University of Padua - Artificial Intelligence

Padua, Italy

MSc IN COMPUTER ENGINEERING

Sep. 2022 - Close to finish

- Weighted grades' mean of 29.27/30, corresponding to a **GPA of 4.0**.
- Hands on experience in **Machine Learning** frameworks and concepts. Fine-tuning of deep learning models for domain specific purposes in **PyTorch** and **TensorFlow** (Computer Vision, Natural Language Processing).
- Further refined my programming skills with **C++**, Python and **Java** projects. Gained knowledge and experience in Big Data frameworks such as Apache Spark (Java) and image manipulation (OpenCV).
- Took part in several development team projects, also **led a team of 10** colleagues to achieve maximum exam grade on a real business case.

University of Padua

Padua, Italy

BSc IN COMPUTER ENGINEERING

Sep. 2019 - Jul. 2022

- Final grade: 110/110 with honors, GPA 4.0 and **top 3%** of my course of study.
- Experimental thesis about Integer Linear Programming exploring alternatives to SotA models.
- Solid foundation in computer science with hands-on experience in full-stack **web development** (React, JS, Express, NodeJS, Google Cloud) and more traditional languages (C++, Java, C).
- Strong academic background, excelling in CS courses as well as key engineering subjects.

Projects

Occluded people image detector

PYTHON, FLASK, C++, HTML5, CSS, PYTORCH, COMPUTER VISION (CV)

Fine-tuned a deep learning model to accurately segment occluded bodies on a dataset of 40K images using distributed computing, achieving over 97% precision on the test set. [Code | Demo | Report]

Gym slot automated booker

JAVASCRIPT, REACTJS, HTML5, FIREBASE, GCP

Backend and frontend application to completely automate bookings for my gym. Got a 12x speedup factor. [Code | Demo]

My own topic-specific Chatbot

PYTHON, LLM, HUGGING FACE, PYTORCH, NATURAL LANGUAGE PROCESSING (NLP)

Created a custom topic-specific dataset and used it to develop my own LLM chat assistant. [Code]

AI Playing Snake

PYTHON, FLASK, HTML5, CSS, TENSORFLOW, REINFORCEMENT LEARNING (RL)

Implemented a Deep Reinforcement Learning Agent from scratch to play the Snake Game. By training it over 1M samples, the agent's performances increased by 20% with respect to the baseline. [Code | Demo | Report]

LLM Code Evaluation

PYTHON, LLM, NATURAL LANGUAGE PROCESSING (NLP)

Developed an evaluation framework for LLM code completion to assess model performance through advanced metrics. [Code | Report]

Volunteering

Croce Verde Padova

Padua, Italy

TEAMWORK, COMMUNICATION, LEADERSHIP, WORKING UNDER PRESSURE

Fed. 2022 - Present

Taken part consistently in emergency ambulance service in my town, dealing with high stress environments and situations.